# Surface Damage Identification for Heritage Site Protection: A Mobile Crowd-sensing Solution Based on Deep Learning

SAFIA MEKLATI and KENZA BOUSSORA, École Polytechnique d'Architecture et d'Urbanisme, Algiers
MOHAMED EL HAFEDH ABDI, Centre d'Imagerie Scintigraphique de Blida, Algeria
SID-AHMED BERRANI, École Nationale Polytechnique, Algeria

This article addresses the general problem of built heritage protection against both deterioration and loss. To continuously monitor and update the structural health status, a crowd-sensing solution based on powerful and automatic deep learning technique is proposed. The aim of this solution is to get rid of the limitations of manual and visual damage detection methods that are costly and time-consuming. Instead, automatic visual inspection for damage detection on walls is efficiently and effectively performed using an embedded Convolutional Neural Network (CNN). This CNN detects the most frequent types of surface damage on wall photos. The study has been conducted in the Kasbah of Algiers where the four following types of damages have been considered: Efflorescence, spall, crack, and mold. The CNN is designed and trained to be integrated into a mobile application for a participatory crowd-sensing solution. The application should be widely and freely deployed, so any user can take a picture of a suspected damaged wall and get an instant and automatic diagnosis through the embedded CNN. In this context, we have chosen MobileNetV2 with a transfer learning approach. A set of real images have been collected and manually annotated and have been used for training, validation, and test. Extensive experiments have been conducted to assess the efficiency and the effectiveness of the proposed solution, using a 5-fold cross-validation procedure. Obtained results show in particular a mean weighted average **precision** of **0.868** ± 0.00862 (with a 99% of confidence level) and a mean weighted average **recall** of **0.84** ± 0.00729 (with a 99% of confidence level). To evaluate the performance of MobileNetV2 as a feature extractor, we conducted a comparative study with other small backbones. Further analysis of CNN activation using Grad-Cam has also been done. Obtained results show that our method remains effective even when using a small network and medium- to low-resolution images. MobileNetV2-based CNN size is smaller, and computational cost better, compared to the other CNNs, with similar performance results. Finally, detected surface damages have also been plotted on a geographic map, giving a global view of their distribution.

CCS Concepts: • **Computing methodologies** → **Neural networks**; *Cross-validation*; Visual inspection;

Additional Key Words and Phrases: Built heritage, surface damage classification, structural health monitoring, deep learning, CNN, data augmentation, small dataset, multi-label classification, K-fold cross-validation, crowd-sensing, GRAD-CAM, geographic mapping

Author affiliation update: S. Meklati and K. Boussora, Laboratoire Ville, Architecture et Patrimoine (LVAP), École Polytechnique d'Architecture et d'Urbanisme, El Harrach, Algiers.
Authors' addresses: S. Meklati and K. Boussora, École Polytechnique d'Architecture et d'Urbanisme, Route de Beaulieu, B.P. n° 177, El Harrach, Algiers, Algeria, 16200; emails: {a.s.meklati, k.boussora}@epau-alger.edu.dz; M. El Hafedh Abdi, Centre d'Imagerie Scintigraphique de Blida, Cité 17 juin, Zabana, Blida, Blida, Algeria, 09000; email: dr.abdi.m.h@scintigraphies.com; S.-A. Berrani, École Nationale Polytechnique, 10 rue des frères Oudek, El Harrach, Algiers, Algeria, 16200; email: sidahmed.berrani@g.enp.edu.dz.
Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2023 Association for Computing Machinery.
1556-4673/2023/03-ART25 $15.00
https://doi.org/10.1145/3569093

**ACM Reference format:**
Safia Meklati, Kenza Boussora, Mohamed El Hafedh Abdi, and Sid-Ahmed Berrani. 2023. Surface Damage Identification for Heritage Site Protection: A Mobile Crowd-sensing Solution Based on Deep Learning. *ACM J. Comput. Cult. Herit.* 16, 2, Article 25 (March 2023), 24 pages.
https://doi.org/10.1145/3569093

## 1 INTRODUCTION

The Kasbah of Algiers is a world heritage city that is over six centuries old. It was built using ancient construction techniques. The UNESCO World Heritage Center and the Advisory Bodies concluded in 2021 that *"the progressive degradation of buildings and housing units remains a concern"* [19]. Today, the survival of the Kasbah of Algiers depends on the preservation and restoration of the remaining buildings, because, there have been *"several collapses of housing units between April 2019 and June 2020, in some cases with tragic consequences for their occupants"* [19]. The UNESCO 2021 report states many factors responsible for this situation: Changes in the traditional ways of life and knowledge system, **erosion** and **siltation**/deposition, natural erosion, lack of coordination of actions, lack of maintenance of dwelling places, land conversion, change in the land use plan, the loss of traditional conservation techniques... This continuous process of degradation changes the state of conservation of the city every day and requires continuous structural health monitoring for its maintenance.

To date, city departments perform periodic condition assessments based on visual and/or technical inspection methods. These traditional methods are certainly effective, but they are often laborious, costly, and time-consuming. They require to have an expert onsite to perform the inspection, and this has to be done periodically.

Alternative solutions rely on the use of automatic inspection techniques that may use detection and classification of damages directly from images of the walls of buildings. In this context, machine learning techniques may be very effective. In particular, deep learning techniques achieve very high detection and classification rates with impressive generalization capabilities for similar tasks [47–49]. These techniques have also been recently used in structural health monitoring with very promising results [48, 50]. Nevertheless, the use of deep learning-based solutions in this context faces many problems and difficulties. To be successful, these solutions require both a huge amount of training data and powerful machines to perform training and detection/classification. Image data has to be collected onsite again, which is in itself costly. Moreover, if experts are in charge of collecting images, then automatic detection/classification techniques lose most of their interest.

In this article, we focus on structural health monitoring and we propose a novel approach that addresses both limitations of traditional approaches and machine learning-based solutions. It also sketches an end-to-end process that makes use of *crowd-sensing* to scale up while highly reducing the cost of the monitoring.

The proposed solution attempts to solve some of the problems associated with the conventional approach, such as:

- Delayed diagnosis of surface damages that could intensify their deterioration or expansion to other intact surfaces.
- The delay in diagnosis leads to more difficult and costly repairs and requires more skilled laborers, advanced equipment, and larger damage inspection projects.
- Large onsite inspection projects may cause disagreement and inconvenience to residents and potential tourists.

By using machine learning techniques, the cost of structural health monitoring is drastically reduced as the process becomes almost fully automatic [30]. Crowd-sensing allows to massively collect wall images and to decentralize damage detection on walls. The user who can be a simple resident is able to take a photo of a suspicious damage and to instantly get a diagnosis. Automatic detection is performed on the smartphone of the user, which saves considerable computation power. The choice of participatory crowd-sensing is also motivated

by the importance of involving the inhabitants, as recommended by the UNESCO [19], and by the profitability of the deployment of the solution (e.g., no need to install specific cameras). Another reason for this choice is that residents are most likely to notice any new surface damage inside and outside their houses.

This schema, however, raises many technical challenges. Training data is required for the machine learning technique, and collecting and annotating this data is very costly. To get rid of the computation power needed for a centralized solution, the diagnosis of wall images has to be done using the computation power of users smartphones. In this case, the machine learning technique that is a deep neural network in our study must be embedded in a mobile application, which requires to drastically optimize the underlying model.

The proposed technical solution relies more specifically on the use of a **Convolutional Neural Network (CNN)** within a transfer learning procedure. The objective is to reduce the number of required training images. The choice of CNNs is motivated by their performance, cost-effectiveness, ease of access, and ability to be retrained on new small datasets (using transfer learning and fine-tuning). In particular, the CNN model MobileNetV2 [39] is chosen to meet the requirements of embedding and running the model on a smartphone.

The article is organized as follows: Section 2 presents existing and related works. Section 3 details the purpose. Section 4 is divided into two parts: Section 4.1 describes how the training and test datasets have been collected and labelled, and Section 4.2 is about the used CNN and its parameters. Section 5 presents the experiments, a comparative study with respect to other small CNNs, a deep analysis of the model, the obtained results, the geographic mapping, and discusses limitations of the present work. Section 6 concludes the article and proposes ideas for future works and possible extensions.

## 2 RELATED WORK

Machine learning techniques have been recently widely used in the domain of cultural heritage [13]. Most of the previous work based on machine learning techniques for structural health monitoring of heritage buildings is summarized in a review article by M. Mishra [30]. In particular, in the specific context of surface damage detection in heritage sites, several techniques have already been proposed [10, 18, 26, 33, 47–50, 55, 56].

Based on the type of input features, existing works for surface damage identification in heritage sites can be classified into two main approaches: The techniques that are based on visual features (using images as input) and those that are not (using any other features). A survey of some of existing works is presented in the following subsections.

### 2.1 Techniques Based on Non-visual Features

Y. Du et al. [10] have proposed a damage assessment of the earthen sites of the Ming Great Wall. A comparison between the **support vector machine (SVM)** and a back-propagation neural network has been performed. The input features to the model consisted of nine evaluation indices (compressive strength, disintegration speed, etc.). The authors consider four types of damage, namely: "collapse," "gully," "sapping," and "crack." They have concluded that SVM is more suitable for this problem. The proposed neural network has only one hidden layer with 18 neurons. K. Zorlu [55] and M. Heidari et al. [18] have both proposed an evaluation of stone weathering using a fuzzy inference system, respectively, in the ancient city of Olba (southern Turkey) and in Persepolis (Iran). In Reference [55], Schmidt hardness and fractal dimension have been used as input features to a fuzzy inference system to predict block weathering. Whereas in Reference [18], changes in physical and mechanical properties such as porosity, and P-wave velocity have been used as input features. A cubic root damage index for weathering prediction have been proposed. The study shows that fuzzy models are reliable for quantifying the degree of weathering of Persepolis.

These techniques rely on machine learning for damage detection in heritage sites. They are very useful but they require to input indices that only experts can measure and provide. Therefore, these techniques cannot be generalized and used within crowd-sensing approach.

## 2.2 Techniques Based on Visual Features

The following two subsections describe the existing techniques that rely on visual features for surface damage detection. These techniques are or could be applied to a built heritage.

*2.2.1 Single Type Damage Detection.* P. Kumar et al. [26] have evaluated several machine learning techniques, as well as several deep learning models (CNNs). The objective is to find damaged cultural heritage photos on the internet to be reviewed by cultural heritage professionals. The dataset for the training has been collected from the internet using Google photos search engine. The proposed models can differentiate heritage from non-heritage sites. Two types of models have been tested. The first type has been trained on images searched without damage keywords. The second model has been trained on all collected images (with and without damage). This was done to determine whether the presence of a damage has an impact or not on the performance of the model, that is, to determine if an image is from a heritage site or not. Both models show satisfactory and comparable results, but with some limitations, in particular in the case of *overlapping* heritage and non-heritage buildings. Authors have also evaluated the accuracy of another model for damage detection, but without specifying the type of damage. The evaluation has been performed using a damage assessment model by Alam et al. [3] on a dataset from the 2015 Nepal earthquake (Social Media for Emergency Relief and Preparedness; SMERP Workshop Dataset [31]). The model, which behaves like a binary classifier, achieves an accuracy of 0.9 (F1-score of 0.84 for the "Damage" class and 0.93 for the "No damage" class).

Other studies based on machine learning for damage detection, either in the heritage [33] or non-heritage context [7, 11, 15, 35], have focused on the detection of cracks. Early studies focused on hand-crafted features (such as edge detection) and later evolved into automatic feature extractors using machine learning and in particular deep learning models.

Perumal et al. [33] proposed an image processing technique for the analysis of monument degradation. Two types of damage are considered: moss and crack. Moss is detected using integrated clustering, and crack is detected using integrated canny edge detection. Although two types of damage are analyzed, they are detected separately by two different techniques. Pragalath et al. [35] also proposed an image processing algorithm with a fuzzy set framework to perform automated evaluation of structural anomalies such as cracks.

Cha et al. [7] proposed a CNN-based method for the detection of cracks. The CNN is compared with a method based on Canny and Sobel edge detection that has been proposed in Reference [1]. The proposed method showed very good performance compared to the edge detection methods. However, it is limited to the detection of a single type of damage (crack).

Gibb et al. [15] have considered the CNN-based model proposed in Reference [7] as a baseline. The network, trained at the baseline on 60,000 images, yields an accuracy of about 0.97, while when trained on a set of 4,880 images, this value drops to 0.88. In Reference [15], a genetic algorithm is proposed for the CNN structure optimization that allows to improve the accuracy to 0.94 with only 4,880 training images, which shows the ability of the CNN to achieve good performance even with relatively small training sets.

Dung et al. [11] proposed a method for crack detection. It is based on a deep fully convolutional network (an auto-encoder like architecture; composed from an encoder and a decoder). Three pre-trained networks have been used and compared as a backbone for the encoder part. A CNN has been trained on 40,000 images for the classification task as a first step, then on 600 labeled crack images for the segmentation task (after adding a decoder part). Results for the classification task (crack/no crack binary classification) vary w.r.t. the used CNN: VGG16 [42] and InceptionV3-based [44] models achieve almost 0.999 of accuracy, while ResNet-based model [16] reaches 0.975 at maximum. Results for the segmentation task based on VGG16 were around 0.9 of precision as well as F1-score [11].

Another type of damage detection, based on a segmentation approach, has been studied by Wang et al. [50]. The study has been conducted on images of roof tiles from the Palace Museum in China. An initial model for detecting and cropping two types of tiles is based on a **Faster region-based convolutional neural network**

**(Faster R-CNN)** [37]. It has achieved an average precision of around 0.91 and 0.890, respectively, for the first and second type of tiles. A second model based on a Mask R-CNN [17] has been trained on the cropped images, for damage segmentation. It has achieved an average precision of 0.975. Only one type of damage was studied in this work, namely, "spalling."

A similar approach based on object detection has been proposed by Zou et al. [56]. A Faster-RCNN based on ResNet 101 [16] has been used for the detection of missing decorative components in the roof of the Forbidden City in China. In this study, two types of missing components have been separately studied. Each model can detect only one type of missing component at a time.

Wang et al. [48] proposed a damage assessment for cultural heritage conservation based on Mobile Crowdsensing and deep learning. The proposed system is composed of three parts, a "mobile client" for data collection, a "web platform" for storage, and a "computer terminal" for damage detection. A Faster Region-based Convolutional Neural Network based on **Zeiler and Fergus (ZF)**-net [52] is used for the detection of damaged bricks in the Great Wall (China). The dataset included four types of damage, which are "Spalling," "Serious spalling," "Efflorescence," and "Crack." However, the model was trained for the detection of damaged bricks, without specifying which type of damage. The most common damage found in the studied part of the Great Wall is spalling, followed by severe spalling. The model achieved an average precision of 0.763 and 0.782 of accuracy.

*2.2.2 Multiple Types Damage Detection.* A first set of works on multiple type damage detection has been proposed in the context of monitoring of the structural health of civil infrastructure (bridges, tunnels, highways, railroads, steel, concrete buildings...). Cha et al. [8] propose a simultaneous detection of multiple types of damages using a Faster R-CNN for structural visual inspection method (bridges and buildings). The proposed solution can detect five types of damages: concrete crack, steel corrosion with two levels (medium and high), bolt corrosion, and steel delamination. Obtained results are 90.6%, 83.4%, 82.1%, 98.1%, and 84.7% average precision ratings for the five damage types, respectively. Li et al. [29] proposes a **Fully Convolutional Network (FCN)**-based multiple damages detection method for concrete structures. A dataset of 2,750 images is used, including four types of damage (crack, spalling, efflorescence, and hole). Results show 98.61% pixel accuracy, 91.59% mean pixel accuracy, for a segmentation approach based on DenseNet-121 [21].

While these works have proved their usefulness for detecting multiple types of non-heritage damage, their application to the heritage context is not certain. For instance, in Reference [8] the studied types of damage are not common to heritage sites. And while some of these types of damage can be found in the heritage context, the visual characteristics may differ, depending on the used materials. The same observation can be made with respect to the method proposed in Reference [29]. An efflorescence damage from a concrete wall has an aspect of salt deposition, while the same type of damage has a different aspect in old buildings with lime-plastered walls. Because the aspect of damage is specific to the structure being studied, the trained model is less likely to be accurate on other types of structures.

For the specific context of heritage, very few works have considered the detection of multiple types of damages. The first work that investigated the simultaneous detection of multiple types of damages using a CNN at a historic site is the one proposed by Wang et al. [47]. Three types of damages are detected: "Crack," "Efflorescence," and "Spall," with an accuracy of 0.943 [47]. Two types of pretrained CNNs have been used: GoogLeNet[43] and AlexNet [24], a small and a medium-size CNN, respectively [5]. The two models have been trained on two different-sized datasets containing 2,000 and 20,000 images [47].

In another work, Wang et al. [49] proposed a Faster R-CNN model based on ResNet101 [16] framework (medium-size CNN [5]) to detect two types of damage: "Efflorescence" and "Spalling." It has been applied for historic masonry structures and has achieved 0.950 of mean average precision. The proposed method makes use of a smartphone or an IP webcam for real time detection by connecting them to a workstation through a Wireless Local Area Network.

Table 1. Some of the Main Machine Learning-based Techniques that Rely on Visual Features for Damage Detection

| Ref. | Publication year | Heritage location | Dataset size | Type of damages | ML technique | Metrics |
|---|---|---|---|---|---|---|
| [7] | 2017 | / | 40,000 | 1 | CNN (S)[1] | Acc. : 97% |
| [11] | 2018 | / | 40,000 | 1 | CNN (S&M&B)[1] | AP: 89.3% |
| [15] | 2018 | / | 3,000 | 1 | CNN (variable) | Acc.: 94% |
| [26] | 2020 | Several + Nepal | 6,075 | /[2] | SVM/LR/RFC/ CNN (S&M&B)[1] | Precision: 0.81/0.94 |
| [50] | 2019 | China | 500 | 1 | CNN (S&M)[1] | AP: 0.975 |
| [33] | 2020 | India | / | 1+1[2] | K-means clustering | MSE : 1,882 |
| [56] | 2019 | China | 278 | 1+1[2] | CNN (M)[1] | mAP: 0.8265 |
| [48] | 2019 | China | 610 | 1+1+1+1[2] | CNN (S)[1] | AP: 0.763 |
| [47] | 2018 | China | 20,000 | 3 | CNN (S&M)[1] | Acc.: 94.3% |
| [49] | 2019 | China | 500 | 2 | CNN (M)[1] | AP: 0.913 |

ML: Machine Learning, MSE: Mean Squared Error, NN: Neural Network, SVM: Support Vector Machine, LR: Logistic Regression, RFC: Random Forrest Classifier, Acc.: Accuracy, AP: Average Precision, mAP: mean Average Precision, S: Small, M: Medium, B: Big.
*1: CNN size was estimated based on the work by Bianco et al. [5]. 2: Several types of damage are included but diagnosed separately or diagnosed as a single class by the proposed model.*

Table 1 provides a summary of the main techniques that rely on visual features for surface damage detection. We can notice that different sizes of datasets and different CNNs are used. Even on relatively small sets, the CNNs are able to achieve good results. Furthermore, the choice of relatively small CNNs did not compromise their effectiveness.

## 3 PURPOSE

In this article, the proposed method is a continuous process based on participatory crowd-sensing approach for automatic multilabel surface damage classification using a deep learning model. As damage detection is based on photos taken using mobile phones, the damage information can later be plotted on a geographical map using location information of photos. This allows cultural heritage professionals and experts to perform a global analysis of how damages spread and possibly to analyze the reasons. The proposed method relies on a small but very effective embedded CNN. The objective is to integrate the CNN into a mobile application that helps citizens to accurately frame the damage without need to have any prior knowledge in the field of architecture and heritage conservation.

The present contribution technically differs from existing works for multiple types damage detection [47–49] by the following main characteristics:

- Damage detection is performed locally on the user smartphone (using an embedded CNN), and not in a centralized manner, which drastically reduces the cost of implementing and using the solution in practice. It also enables a participatory crowd-sensing approach at a lower cost.

- A classification approach is used (instead of detection or semantic segmentation); the model tells if a damage is present in a photo or not and does not have to provide a bounding box or a segmentation map. This approach significantly reduces the computational cost of the solution. It avoids the use of two models (one for detection and one for classification), and also the use of a segmentation head, which contains more layers (and hence more trainable parameters and more computations) than a simple classification head.
- The model addresses simultaneously several damages within a multi-label classification approach, which is very suitable to address the case of overlapping damages in the same wall (and then in the same photo).

Our proposed solution is presented following the four following parts:

(1) Collecting and labeling a dataset, from the Kasbah of Algiers (for the first time, to the best of our knowledge). Part of the dataset is made available online for researchers following this URL: https://data.mendeley.com/datasets/zwshr65gnc.
(2) Training a CNN, adapted to mobile deployment (to be embedded in a mobile application).
(3) Assessing of the CNN effectiveness.
(4) Plotting damage information on a geographic map.

## 4 MATERIALS AND METHODS

### 4.1 Datasets

*4.1.1 Image Characteristics.* Wall photos have been collected from the Kasbah of Algiers, over six different days, outside houses, after obtaining the required authorization. To the best of our knowledge, this is the first time that such a dataset for surface damage detection has been collected at this specific site. Two mid-range phones have been used. The photos have been taken in daylight, but at different times.

The images used in our study (for training/validation and for the first step of testing) have been cropped from the original images.[1] The objective is to have in each image only the suspicious surface. As a result, the resolution of the cropped images is variable. However, some of the cropped images contain multiple types of damage that are superimposed. These images are very useful to assess the ability of the CNN to deal with this *multi-label* classification problem. This is particularly important, as the presence of multiple overlapping damages in a same wall is frequent in the Kasbah of Algiers.

A total number of 2,674 images have been collected using two smartphone cameras: (1) A single 12 Megapixel camera, with integrated localization information (A-GPS GLONASS, BDS) and (2) a 13 Megapixel single camera. The resolution of resulting cropped images ranges from $224 \times 224$ to $1{,}500 \times 1{,}500$ pixels. This dataset have been divided into two subsets: A training/validation set of 2,040 images and a test set of 634 images (we refer to it as *Test1* in what follows).

An additional set of images has been collected (we refer to it as *Test2* in what follows), with the purpose of evaluating the performance of the CNN on images taken at different distances from the wall and from different angles. A total of 142 images have been collected. These images have not been cropped as in the previous sets (training/validation set and Test1). A total of 25 walls with damage have been photographed. Some of these walls contain only one type of damage, while others are in the form of a complex combination of two or three damages. Each wall has been photographed 5 to 6 times, with 2 to 3 variations in distance (near: <1 meter; middle: 1–2 meters; and far: >3 meters) and two variations in terms of angle of view (approximately 45 degrees to the wall, to the right and to the left).

In this context, labeling consists in associating the identified damages with each image by an expert. As explained before, several damages can be associated with the same image. Considered damages are the following: Efflorescence, Spall, Crack, and Mold.

[1] In practice, users can also select the suspicious region of the wall to focus on. Therefore, this cannot be considered as a strong limitation or a constraint.

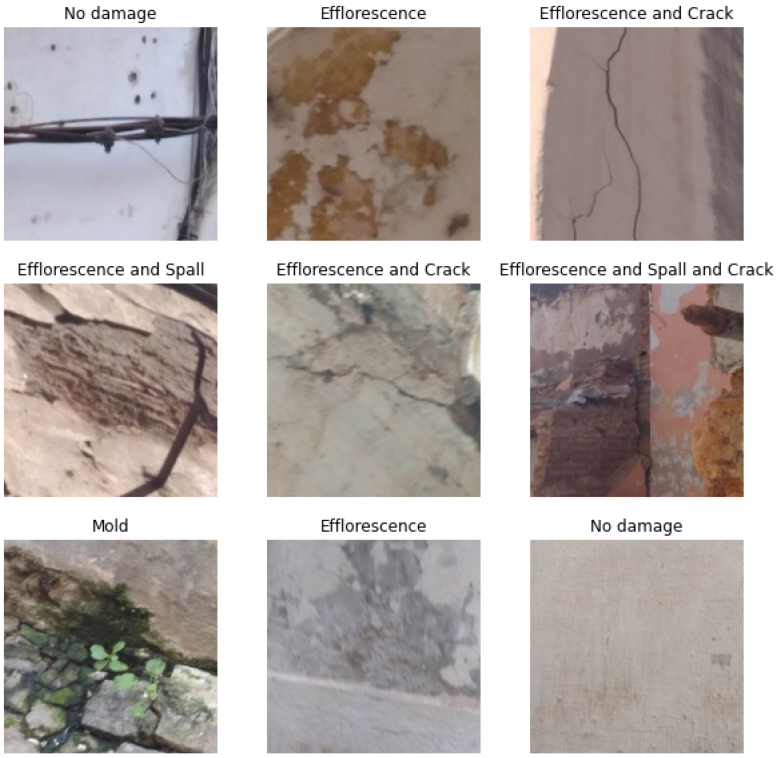



Fig. 1. A sample of collected images with their respective labels.

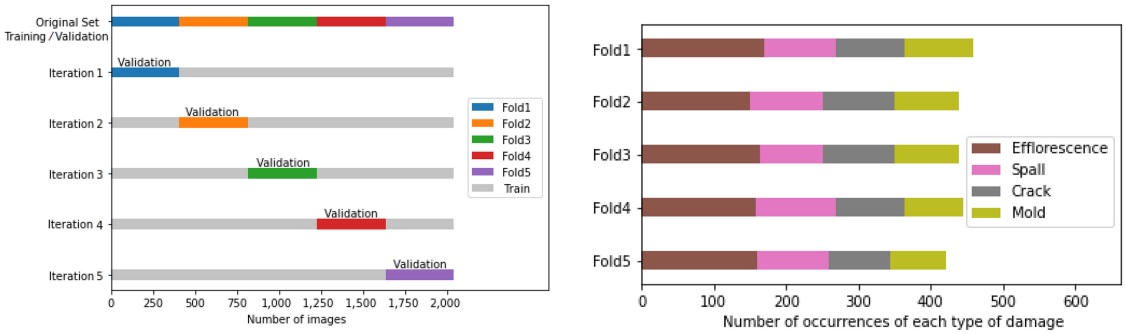
Table 2. Description of the Training/validation and Test Sets

| | Original Training/Validation Set | Test1 Set | Test2 Set |
|---|---|---|---|
| Efflorescence | 799 (39.16%) | 249 (39.27%) | 64 (45.07%) |
| Spall | 495 (24.26%) | 123 (16.40%) | 47 (33.09%) |
| Crack | 474 (23.23%) | 143 (22.55%) | 48 (33.80%) |
| Mold | 432 (21.17%) | 116 (18.29%) | 38 (26.76%) |
| No damage | 356 (17.45%) | 142 (22.39%) | / |

Examples of collected images with their respective labels are illustrated in Figure 1. Details on the training/validation and test sets are given in Table 2.

Finally, all images from both training and test sets have been resized to $224 \times 224$ pixels.

*4.1.2 Preparing the Training/validation Set.* Given the relatively small size of the dataset, a K-fold cross-validation has been applied. This procedure is generally used to estimate the performance of a learning algorithm on new data, and in particular to detect overfitting or selection bias problems [6], which could be present when using small sets. To apply this procedure, the data is first divided into K folds of almost equal sizes. Then, K training and validation iterations are performed. In each iteration, a different fold of the data is retained for validation while the other folds are used for training. The data are usually stratified before being divided into K folds. Stratification involves rearranging the data so each fold is a good representative of the whole set. For




(a) The preparation of the dataset for 5-fold Cross-validation.

(b) The number of occurrences of each label in each fold.

Fig. 2. The preparation of the training/validation set for cross-validation and occurrences of each label in the five folds.

example, in a binary classification problem where each class represents 50% of the data, it is better to organize the data so in each fold, each class represents about half of the instances [36]. Performance indicators can be calculated, such as accuracy, precision, recall, or F1-score. In our study, we have used a 5-fold cross-validation. This choice is motivated by the computational cost. Although it is recommended to choose a larger K, the upward bias cannot be neglected even in 5-fold or 10-fold cross-validation [14].

After applying a random shuffle on the training dataset, it has been divided into five folds, with 408 pictures in each fold (see Figure 2(a)). The number of occurrences of each label in these five folds is illustrated in Figure 2(b). It can be noticed that the number of occurrences of each damage (or label) in the five folds has a globally comparable distribution. At each training iteration, a different fold has been held for validation and the other folds for training. Illustration of the procedure is given in Figure 2. For example, for the first iteration, the first fold has been used for validation and the other four folds for training. For the second iteration, the second fold has been used for validation and the other four folds are used for training and so on.

Also, as the number of images is relatively low, to avoid overfitting, data augmentation has been applied. Overfitting refers to the phenomenon in which a network learns a function with high variance so it models the training data perfectly and is less able to generalize to unseen data (validation or test sets). Data augmentation encompasses a set of techniques that improve the size and quality of learning datasets [41].

In this work, data augmentation has been applied using a sequential model (integrated as a layer in the global model) with the following transformations: horizontal and vertical random flip, random rotation of 0.05, random zoom of 0.05 factor in height and width, random translation also of 0.05 factor in height and width, and random contract of 0.2. An example of an augmented image is illustrated in Figure 3.

## 4.2 The Convolutional Neural Network

CNN architectures have evolved since the first use of mini-batch gradient descent (based on back-propagation) by Y. LeCun et al. [28] for the recognition of hand Handwritten digits. Krizhevsky et al. trained a CNN (AlexNet) [24] on 1.2 million high-resolution images from the ImageNet LSVRC-2010 [38] contest and achieved top-1 and top-5 error rates, respectively, of 37.5% and 17.0%. Subsequently, increasing the depth of the CNN (VGG) [42] showed a significant improvement over previous configurations. Another evolution has been the use of **residual networks (ResNet)** [16], which are easier to optimize and can increase the depth of the CNN, while being less complex than the VGG. However, lightweight deep neural networks have been developed for mobile deployment, especially by using depth-wise separable convolutions (MobileNetV1 [20]). As a result, MobileNetV1 is almost as accurate as VGG16 while being 32 times smaller and 27 times less demanding on computing resources [20]. MobileNetV1 has been described as a powerful feature extractor and only requires about 4.9 to 5.7 times fewer

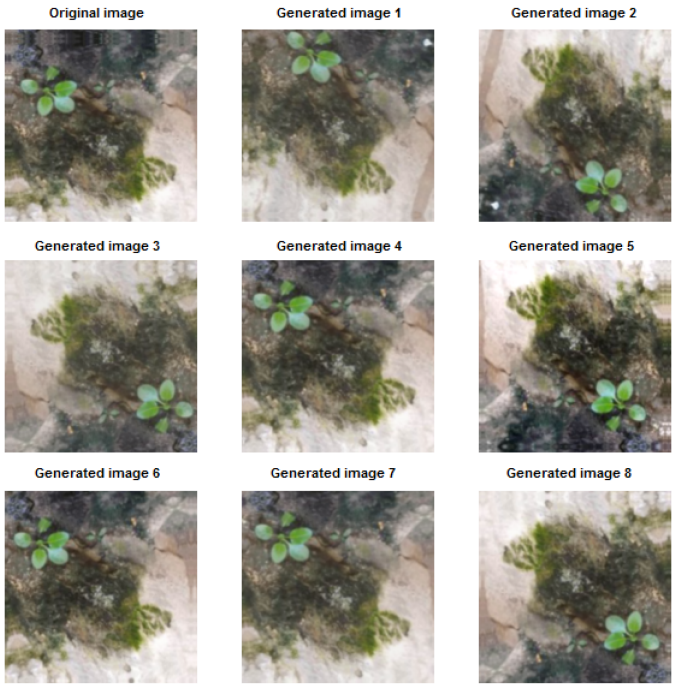



Fig. 3. An illustration of an augmented image.

multiply-adds (computational cost) than ResNet-101 [16, 39]. MobileNetV2 has then been developed to further reduce the computational cost by using ($3 \times 3$) depth-wise separable convolutions [39]. The computational cost is 8 to 9 times smaller than MobileNetV1 at only a small reduction in accuracy. MobileNetV2 architecture is based on an inverted residual structure where the shortcut connections are between the thin bottle-neck layers (in contrast to the case of ResNet, where the shortcut connections are between thick layers). Figure 4 illustrates this architecture. As a result, this network is one of the most efficient when focusing on the top-1 accuracy density, which is calculated by dividing the top-1 accuracy by the number of parameters of the network. This value is almost 20%/M-params for MobileNetV2, around 10 for GoogleNet, and 1 for AlexNet [5].

Additionally, others networks have been developed for faster inference and better parameter efficiency. One of the most efficient small CNNs is EfficientNetV2B0 [45]. Its architecture is based on the replacement of the depthwise conv3×3 and expansion conv1×1 blocks (used in MobileNetV2) by a single regular conv3×3. This CNN has a higher top-1 accuracy than MobileNetV2 on ImageNet [38] dataset (78.7% vs. 71.3%, respectively) [46]. Other small CNNs suitable for mobile deployment are NASNetMobile [54] and DenseNet121 [21].

The verification of the feasibility of the proposed method, based on mobile crowd-sensing, requires the use of infrastructures similar to those that will be used for its deployment. This has motivated the choice of a small neural network, with low computational cost, to be deployed on mid-range phones. The objective is to verify that the characteristics of this range of phones (storage space, RAM, and camera resolution) neither compromise the execution of the CNN nor reduce its efficiency.

These relatively small networks that are pre-trained on millions of images (ImageNet dataset [38]), in particular MobileNetV2, have been used as a feature extractor, using transfer learning in several previous works. Even when trained on a small dataset, the CNN achieved very good performance levels. This approach has been shown to be effective for face recognition [27], driving assistance [4], fruit classification [51], and especially in the medical

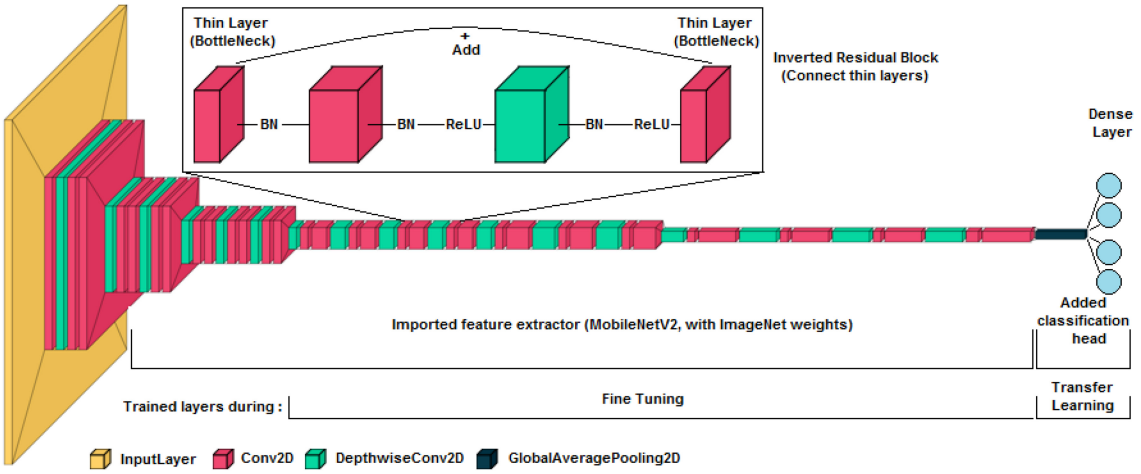



Fig. 4. Visualization of our CNN model architecture, based on MobileNetV2. *BN: BatchNormalization, ReLU: Rectified Linear Unit. For simplification purposes, some layers are not represented, like ZeroPadding2D.*

field on relatively small sets [2, 32]. Given this considerable evolution of CNN architectures intended for mobile deployment, designing a new network will have very little chance of achieving better results.

In our study, based on this review, we have chosen MobileNetV2 because of its small size (14 MB) and low computational cost (time per inference step on CPU of 25.9 ms, and 3.8 ms on GPU), which is suitable for deployment on mobile devices, with a good performance (0.713 top-1 accuracy on the ImageNet validation dataset) [46]. For better evaluation of this backbone as a feature extractor, it has been compared to other small CNNs such as EfficientNetV2B0, one of the most recent and efficient small CNNs, and also to NASNetMobile and DenseNet121.

Each one of these CNNs has been used in this work as a feature extractor, applying a transfer learning approach as a first step and fine-tuning as a second step. Our architecture (for MobileNetV2-based CNN) is depicted in Figure 4.

Using MobileNetV2 as a feature extractor, means that it has been imported without its original classification head (of 1,000 classes). A new classification head has been added (a single dense layer with only 4 classes). As a result, our model architecture is resumed in this feature extractor (with the original weights trained on ImageNet dataset) and the new classification head on top of it.

The configuration of this classification head has been related to the specificity of the heritage site. A single dense layer has been added to have the smallest possible CNN size. Four classes have been used, corresponding to the four most frequent found damages in the Kasbah. However, there were several images of more than one damage type in the same frame, so a *multi-class* classification is deemed inappropriate for Kasbah. A non-hierarchical *multi-label* classification has been evaluated in this article. In this type of classification, the CNN is supposed to detect multiple overlapping damage types in a single image, giving a probability for each damage type independently of the others. For this reason of independence, a Sigmoid activation is used. It has to be noticed that it may be difficult to apply a bounding boxes approach to the context of the Kasbah, as the different types of damage may overlap and the walls of the Kasbah are not made of exposed bricks. In addition, using two CNNs would be less efficient in terms of computational cost and storage space.

The different layers of the used CNN, from the input to the top are as follows:

(1) Input layer with a shape of (3,224,224).
(2) Data augmentation layer: as described in Section 4.1.2.

Table 3. Used Hyper-parameters for Training (for All Tested CNNs)

| **Parameter** | Transfer learning step | Fine-tuning step |
|---|---|---|
| **Input shape** | (3,224,224) | (3,224,224) |
| **Batch size** | 51 | 51 |
| **Weights** | ImageNet | Transfer learning step |
| **Optimizer** | Adam | RMSprop |
| **Learning rate** | $10^{-4}$ | $10^{-5}$ |
| **Loss function** | Binary Cross-entropy | Binary Cross-entropy |
| **Dropout rate** | 0.3 | 0.3 |
| **Output layer (Classifier)** | Sigmoid | Sigmoid |
| **Epochs** | 200 | 25 |

(3) Preprocessing layer: The default preprocessing input function of the MobileNetV2 has been used, converting pixel values form the range of [0, 255] to [−1, 1] for the training of the MobileNetV2-based CNN and the corresponding default preprocessing input function for each one of the other used CNNs.
(4) Feature extractor (based on MobileNetV2) or one of the other CNNs (EfficientNetV2B0, NASNetMobile and DenseNet121).
(5) GlobalAveragePooling2D layer: Converts the features from a block of spatial locations, to a vector (of 1,280 elements for MobileNetV2 and EfficientNetV2B0, 1,056 for NASNetMolbile, and 1,024 for DenseNet121).
(6) DropOut: Randomly sets input units to 0 with a frequency of 0.3, which helps prevent overfitting. A higher value reduces the learning capabilities of the network and slows down the training process, and a lower value does not sufficiently prevent overfitting.
(7) Dense layer: Converts the features into a single prediction per image. A layer of four units (neurons) is used, corresponding to the four types of damage (four classes).
(8) Sigmoid activation layer: Transforms the output values into independent probabilities, which is suitable for a multi-label classification.

A sequential model has been used during the training step. A baseline source code from the Tensor-Flow library with predefined parameters has been used and adapted [9]. This choice has been motivated by the availability of the documentation and the predefined parameters, which have been modified slightly to get good results. A CNN with good parameters is more likely to give good results on small sets. Starting a code from scratch would be more difficult and less likely to give such good results.

During the first step of training (transfer learning step), the model has been compiled using the following parameters: Adam optimizer, with a learning rate of 0.0001, and BinaryCrossentropy as loss function. Metrics has been displaying average precision and each class precision. The number of trainable parameters of the resulting model is equal to 5,124.

Training has been done in the Google Collaboratory environment, using GPU (Tesla T4). During this first step, the training has been realized for 200 epochs. Each epoch took about 4 seconds. The training took about 13 minutes at every cross-validation iteration.

During the second step of training, called fine-tuning, the top layers of the CNN have been unfrozen (starting fine-tuning at the layer 100). The same loss function as the first step has been used, but changing the optimizer to RMSprop and decreasing the learning rate (10 times). Trainable parameters have been 1,866,564 out of 2,263,108 total parameters of the model. The training has been realized for 25 epochs. Beyond this number of epochs, the CNN starts to overfit, as a large number of parameters has to be trained on a relatively small dataset.

The main hyper-parameters with the corresponding values are summarized in Table 3.

# 5 RESULTS AND DISCUSSION

## 5.1 CNN Model Evaluation

Five evaluation types of the trained MobileNetV2-based CNN has been performed, using:

- Metrics (Precision, Recall, F1-Score, and **Area Under the Curve (AUC)** of the **Receiver Operating Characteristic (ROC)** curves).
- A comparative study with respect to other small CNNs.
- Confusion Matrix.
- Grad-Cam.
- Evaluation with respect to damage-free images.
- Evaluation with respect to variations of wall distance and viewing angle.

*5.1.1 Evaluation Using Precision, Recall, and AUC.* Predictions are generated by the CNN as an array of four elements. Each element represents the probability range from 0 to 1 for the corresponding damage, separately. For example, the Efflorescence damage illustrated in Figure 7(a) has obtained the following prediction array: [0.99, 0.14, 0.01, 0]. The default threshold of 0.5 has been used. The threshold is the cut-off value above which a damage is considered as present and below as not-present. The example in Figure 7(a) has been predicted by the CNN as containing only an Efflorescence damage with a probability of 99%. The probability for the other three damage types has been less than the threshold value of 50%. So, the prediction array has been as follows: [1,0,0,0] (identical to the ground truth). The four elements of output for this example have been a true positive for Efflorescence and three true negatives for the rest of damages. **True Positives (TP)** and **True Negatives (TN)** have been then calculated, along with **False Positives (FP)** and **False Negatives (FN)**. These values have been used to calculate precision, recall, and F1-Score for each label.

An average value of each metric of the four classes has been then calculated. Three averaging methods are used; micro, macro, and weighted average.

- micro: Calculate metrics globally by counting the total TP, FN, and FP.
- macro: Calculate metrics for each label and find their unweighted mean. This does not take label imbalance into account.
- weighted: Calculate metrics for each label and find their average, weighted by support (the number of true instances for each label). This alters "macro" to account for label imbalance; it can result in an F-score that is not between precision and recall.

A mean value of the loss, each average metric, as well as per class metrics, of the five iterations has been then calculated, with the standard deviation (99% of confidence level).

After 200 epochs in the first training step (transfer learning), obtained results have been 0.2776 ± 0.0361 of mean loss and 0.7957 ± 0.0231 of mean average precision (calculated mean values ± standard deviation, on the validation folds of the five iterations).

After 25 following epochs during the second training step (fine-tuning), the results have been improved significantly to 0.242 ± 0.0192 of loss and 0.8274 ± 0.00923 of mean average precision (calculated mean values ± standard deviation, with 99% confidence level, on the validation folds of the five iterations).

The obtained results on the test1 set are **0.1966** ± 0.0117 of **loss** and **0.868** ± 0.00862 of mean weighted average **precision** (calculated mean values ± standard deviation, with 99% confidence level).

Detailed metrics of precision recall and F1-score are given in Table 4. It can be noted that overall, the CNN performs well in detecting all four damage types. Even if the precision is a bit lower for the efflorescence damage compared to the other types of damage, the performance remains good. This type of damage is the most frequently found in the Kasbah and the most associated with the other types of damage. Also from Table 4, we can notice that there is no significant difference in the metric values between the 5 iterations, using the 5-fold cross-validation.

Table 4. Detailed Evaluation Results on the Test1 Set, with Mean Values over the Five Iterations (MobileNetV2-based CNN)

| | Precision | Recall | F1-score | Label occurrence |
|---|---|---|---|---|
| Efflorescence | 0.772 ± 0.0286 | 0.828 ± 0.0421 | 0.798 ± 0.00862 | 249 |
| Spall | 0.956 ± 0.0269 | 0.846 ± 0.0441 | 0.896 ± 0.0172 | 123 |
| Crack | 0.874 ± 0.0453 | 0.902 ± 0.0184 | 0.888 ± 0.0184 | 143 |
| Mold | 0.968 ± 0.0321 | 0.79 ± 0.0895 | 0.866 ± 0.0487 | 116 |
| micro avg | 0.858 ± 0.00862 | 0.84 ± 0.00729 | 0.848 ± 0.00461 | 631 |
| macro avg | 0.892 ± 0.0134 | 0.84 ± 0.0126 | 0.862 ± 0.0134 | 631 |
| weighted avg | 0.868 ± 0.00862 | 0.84 ± 0.00729 | 0.852 ± 0.00862 | 631 |

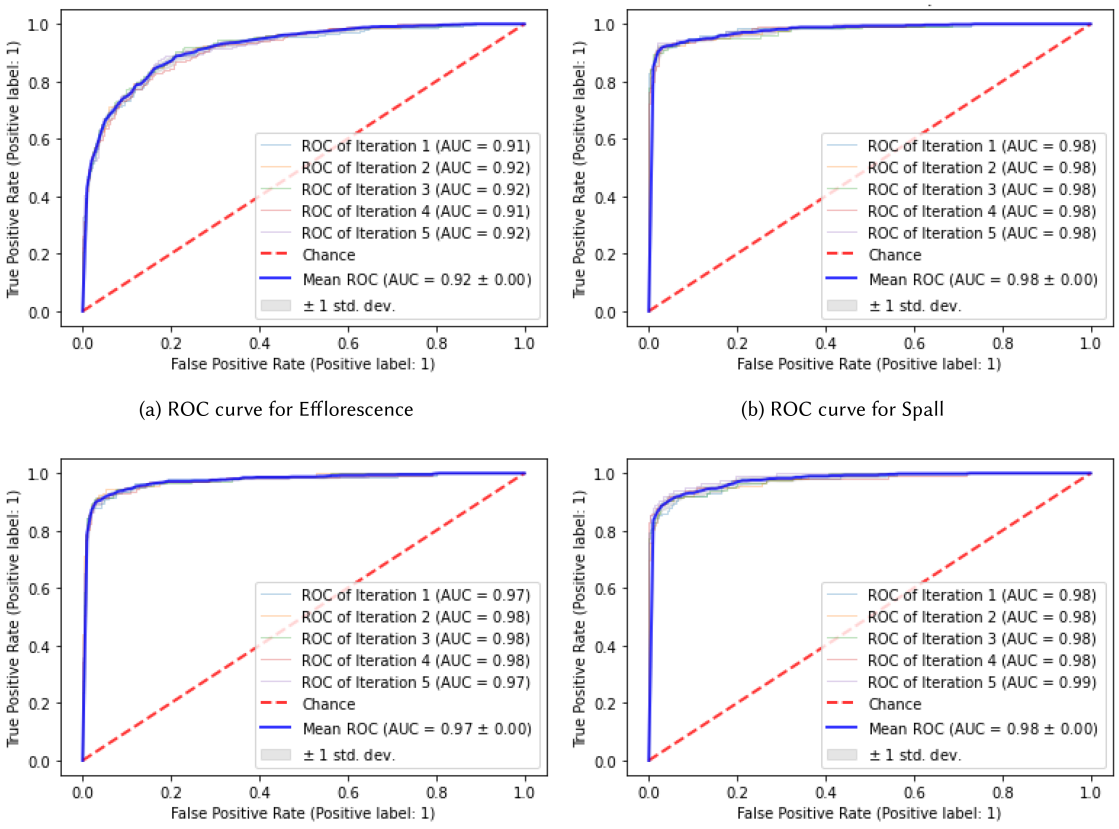
ROC curves are useful to visualize the performance of the classifiers. They are increasingly used in machine learning [12]. They consist of a graphs in which TP rate is plotted on the Y axis and FP rate is plotted on the X axis. Calculating the **Area Under the Curve (AUC)**, is a method to reduce ROC performance to a single scalar value [12]. A perfect classifier would have an AUC of 1 and a random one of 0.5.

ROC curves have been plotted for the four damages in Figure 5. For each damage, the five ROC curves corresponding to the five realized iterations (cross-validation), have been generated, along with their mean curve (strong line in blue color).

From Figure 5, it can be noticed that the AUC, for Spall (Figure 5(b)) and Mold (Figure 5(d)) are the highest, with a value of 0.98, and also for Crack with a value of 0.97 (Figure 5(c)). While, even if the AUC of Efflorescence is the lowest with a value of 0.92 (Figure 5(a)), but it is still a good performance. In fact, Efflorescence damage is the most frequent in the Kasbah of Algiers because of its location by the sea and also the recent installation of water connections with a risk of leakage.Also from Figure 5, it can be noticed that there has been no significant difference between the five iterations in each label (the five iteration curves are almost superimposed for each damage).

*5.1.2 Evaluation with Respect to Other CNNs.* The ROC curves for the the other tested CNNs with different backbones (EfficientNetV2B0, NASNetMobile, and DenseNet121) were almost the same as those of MobileNetV2-based CNN. For that reason, and for simplification purposes, only a table comparing AUC values of these CNNs is illustrated (Table 5). The results show almost similar performance of the different used backbones especially between MobileNetV2 and EfficientNetV2B0-based CNN when using the GPU, both in training and inference time. However, the MobileNetV2-based CNN was *faster* when using the *CPU* for inference, and its size is significantly *smaller* than the other CNNs. However, AUC values were also similar for all test CNNs, except that MobileNetV2 is slightly better on test2 images. The results reinforce the choice of MobileNetV2 as a feature extractor, as it seems more suitable for use in the context of the Algiers Kasbah, as most of the inhabitants are middle to poor class, with medium- to low-range smartphones (limited storage space).

*5.1.3 Evaluation Using Confusion Matrix.* Confusion matrix is a useful and comprehensive presentation of the classifier performance. It is commonly used in the evaluation of multi-class classifiers. Methods of calculation and source codes are widely available for this first case of multi-class approach. However, with multi-label classifiers, the most available calculation methods use a binary confusion matrix for each label, thus it does not give a global representation over all labels. Krstinić et al. [25] have proposed a novel method for the computation of a confusion matrix for multi-label classifiers. In this context, the confusion matrix is computed considering the accumulation of the contribution of each sample based on its match with the ground-truth. For instance, if an image contains two damages, Efflorescence and Crack, and it has been predicted to contain these two damages along with a Spall




(a) ROC curve for Efflorescence

(b) ROC curve for Spall

(c) ROC curve for Crack

(d) ROC curve for Mold

Fig. 5. ROC Curves for the four different labels (results on the test1 set of the MobileNetV2-based CNN).

damage, then the diagonal of Efflorescence and crack will be incremented by 1, and the off-diagonal (misclassified element) of these same damages with respect to Spall by 0.5. In other words, a Spall damage has been confused with an Efflorescence and a Crack by a factor of 0.5 for each one. The confusion matrix is finally normalized to show the precision of each label on the diagonal. An open-source code from Julián D. Arias-Londoño is also available [22]. A similar approach has been used to calculate the confusion matrix for the present five iterations, which are illustrated in Figure 6.

From the confusion matrix of the five models illustrated in Figure 6, we can notice that Efflorescence damages are confused in some cases with Spall and Mold. The following reasons could explain this: Efflorescence has been the most frequent damage and the most associated with other damages (Table 2). This damage may have several forms, making it difficult for labeling by the expert and also for detection by the CNN. It can have several stages, ranging from a border line with a normal aspect to a pre-Spall damage. It has also been associated and overlapping with mold damages. This can explain the lowest performance for this label, in comparison to the other three labels.

*5.1.4 Evaluation Using Grad-Cam.* **Grad-Cam (Gradient-weighted Class Activation Mapping)** is a visual explanations tool of the decisions made by a CNN [40]. It uses the gradients of any class, which are passed to the final convolutional layer to produce a coarse location map highlighting the regions of the image that are important for the prediction of that class. It can be applied without architectural changes or re-training.

Table 5. Comparative Characteristics and Results of the Trained CNNs – Results on Test1 & Test2

| | MobileNetV2 | EfficientNetV2B0 | NASNetMobile | DenseNet121 |
|---|---|---|---|---|
| Backbones | | | | |
| **Size**[46] | **14 MB** | 29 MB | 23 MB | 33 MB |
| Top-1 Accuracy[1] | 71.3% | **78.7%** | 74.4% | 75.0% |
| Parameters[46] | **3.5M** | 7.2M | 5.3M | 8.1M |
| Training | | | | |
| Time (min)[2] | **13** | **13** | 17 | 23 |
| Size (H5)[3] | **16.2 MB** | 42.4 MB | 34.9 MB | 51.6 MB |
| **Size (TFLite)**[4] | **8.47 MB** | 22.4 MB | 16,3 MB | 26,6 MB |
| GPU inference (s)[5] | 3 | **2** | **2** | 3 |
| **CPU inference (s)**[6] | **23** | 30 | 50 | 91 |
| Test1 AUC[7] | | | | |
| Efflorescence | 0.92+/−0.00 | 0.92+/−0.00 | 0.91+/−0.00 | **0.94+/−0.00** |
| Spall | **0.98+/−0.00** | **0.98+/−0.00** | 0.97+/−0.00 | **0.98+/−0.00** |
| Crack | 0.97+/−0.00 | **0.98+/−0.00** | **0.98+/−0.00** | **0.98+/−0.00** |
| Mold | 0.98+/−0.00 | 0.98+/−0.00 | 0.98+/−0.00 | **0.99+/−0.00** |
| Test2 AUC[8] | | | | |
| Efflorescence | **0.88 +/−0.02** | 0.84+/−0.01 | 0.82+/−0.02 | 0.84+/−0.01 |
| Spall | **0.92+/−0.02** | 0.87+/−0.01 | 0.85+/−0.01 | 0.87+/−0.01 |
| Crack | **0.97+/−0.01** | 0.93+/−0.01 | 0.95+/−0.01 | 0.96+/−0.01 |
| Mold | 0.96+/−0.01 | 0.98+/−0.00 | 0.95+/−0.01 | **0.99+/−0.00** |

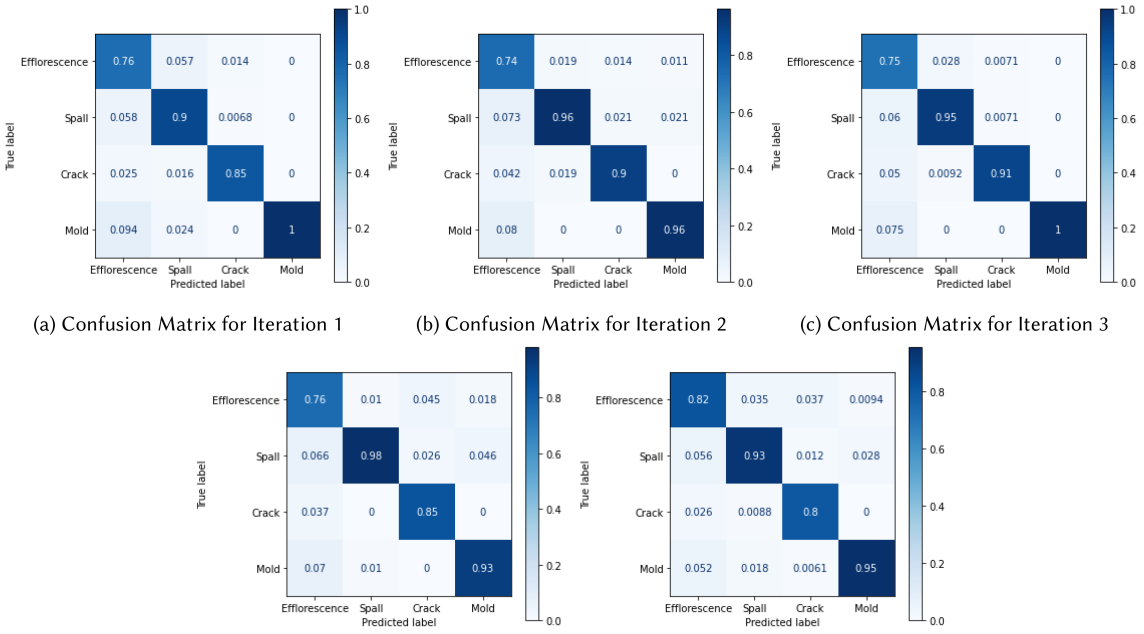
AUC: Area Under the Curve of the Receiver Operating Characteristic curves. *1: Top-1 Accuracy on the validation set of ImageNet [38]) [46]. 2: Training time of each CNN at each iteration using GPU (Tesla T4 1.59GHz) in minutes. 3: H5 format is the standard Keras file format for trained CNN. 4: TFLite format is used for deployment on mobile devices and smartphones. 5: Inference time in seconds on Test1 (364 images), using GPU (Tesla T4 1.59GHz). 6: Inference time in seconds on Test1 (364 images), using CPU (2 x Intel(R) Xeon(R) 2.20 GHz). 7: AUC results of Test1 (634 images). 8: AUC results for Test2 (142 images) containing variations in distance and viewing angle from the walls.*

The CNN of the first iteration has been chosen for this section, and it has been converted from a nested (sequential) form to a flattened form to facilitate the application of Grad-Cam.

The Grad-Cam has been applied on the last convolution layer (*Conv_1 (Conv2D)*), having the output shape of (None, 7, 7, 1,280). The resulting Grad-Cam activation map has a shape of 7 × 7 pixels. This activation map has been then superimposed with the original image (224 × 224 pixels). An illustration of three correctly classified damages with a single label (single damage) is given in Figure 7.

Another example, to demonstrate the ability of the CNN to detect more than one damage and to locate each of them in the same image, is illustrated in Figure 8. In this example, there were two types of damage: a Crack in the middle of the image and Efflorescence around it, especially in the upper part of the image. The ability of the CNN to deal with such situations is important, as they are common in the Kasbah. The image in Figure 8(a) has been correctly predicted by the model as containing two types of damage, Efflorescence and Crack, with a probability of 93% and 100%, respectively. The predicted probabilities for the other two types of damage (Spall and Mold) has been around 0% for each. Two class activation maps for the two predicted damages have been generated and superimposed with the original image (Figure 8(b) and 8(c)). It can be noticed that the CNN has been able to recognize the region of the picture containing the Crack (Figure 8(c)), along with the surrounding area containing Efflorescence (Figure 8(b)).

The analysis of the few misclassified images using the Grad-Cam showed that even if a probability of less than 50% was predicted for a present damage, it remains the highest value compared to the probability of the




(a) Confusion Matrix for Iteration 1 (b) Confusion Matrix for Iteration 2 (c) Confusion Matrix for Iteration 3

(d) Confusion Matrix for Iteration 4 (e) Confusion Matrix for Iteration 5

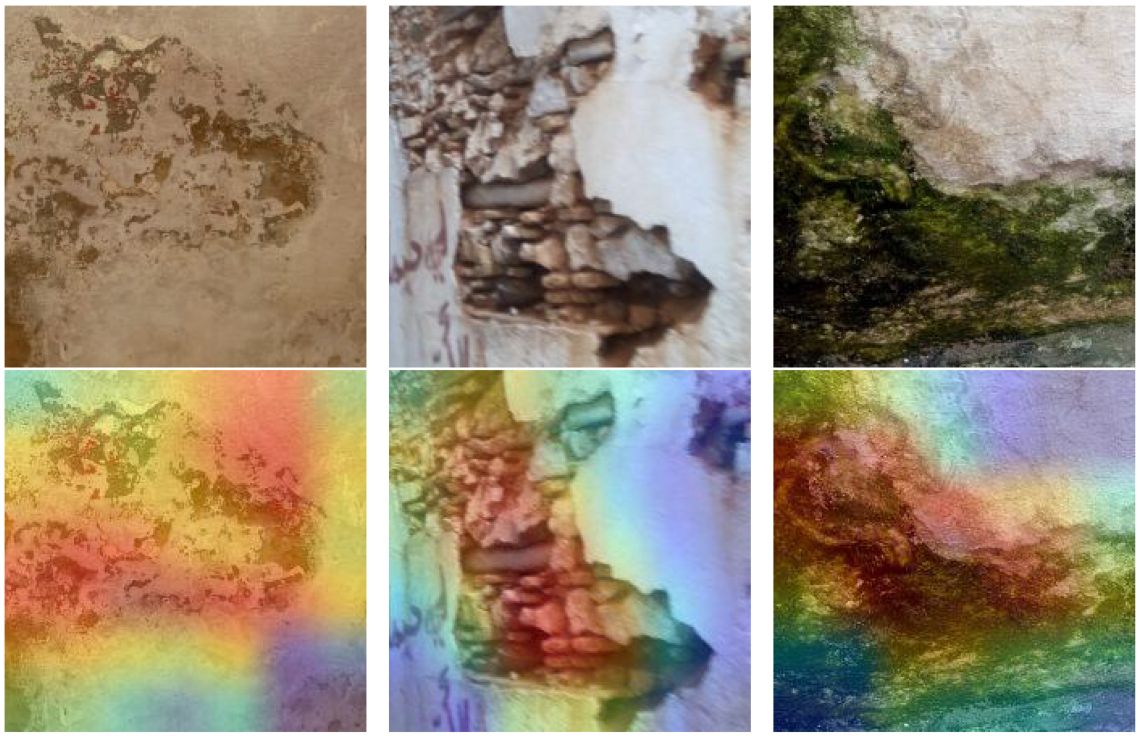
Fig. 6. Confusion Matrix for the five iterations.

other non-present damages. Grad-Cam also showed that the CNN was able to locate approximately the damage in the picture. In some cases, the damages have occupied a small part of the image, which has been probably responsible for a probability lower than 50%; such an example is given in Figure 9. In this example, there has been only one type of damage, which is Efflorescence (Figure 9(a)). The CNN predicted a probability of 42% for this damage and less than 10% for the other three types of damage. This probability is less than the threshold value of 50%, so the prediction of the CNN is considered as no type of damage is present, which is wrong. However, the CNN is still able to recognize the presence of the damage in the picture, even with a low probability, and is able to approximately locate it (Figure 9(b)). Thus, the deployed CNN should be able to help the users to better select the region of the wall with the higher probability of damage.

*5.1.5 Evaluation with Respect to Damage-free Images.* Another analysis has been performed on of subset of images without damages from Test1. The size of this subset is 142 images, which represents 22.39% of the whole size of Test1. The majority of these images has been correctly predicted by the CNN. Some of them have been wrongly classified as follows (mean values ± standard deviation are calculated for the five iterations, with a 99% confidence level)):

- 11.686% ±2.33 as containing an Efflorescence damage,
- 0.175% ±0.39 as containing Spall,
- 8.975% ±3.538 as containing Crack,
- 0.35% ±0.451 as containing Mold.

Obviously, most false positive cases concern the Efflorescence type of damage, which is consistent with the results shown before in the analysis using ROC curves (Figure 5) and the confusion matrix (Figure 6). An example of three damage-free images correctly predicted by the CNN is given in Figure 10(a), 10(b), and 10(c). Another example of an image with no damage, but incorrectly predicted as containing Efflorescence damage, is shown

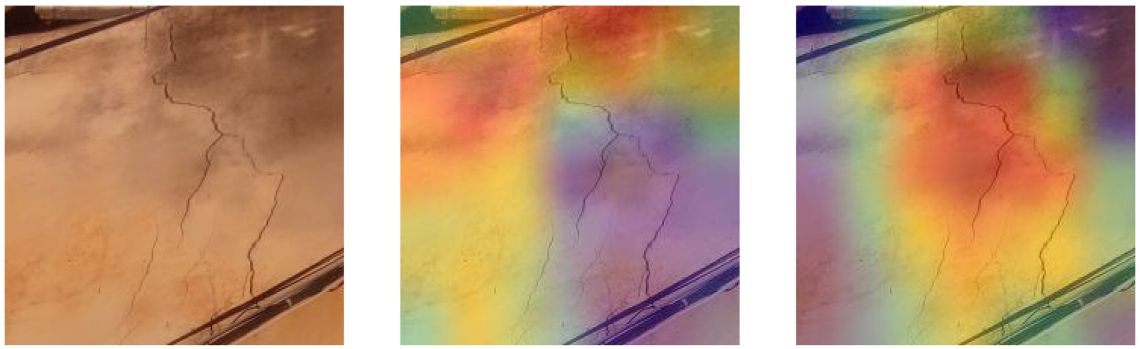


(a) Efflorescence (correctly predicted with 99% probability)

(b) Spall (correctly predicted with 100% probability)

(c) Mold (correctly predicted with 100% probability)

Fig. 7. Three examples of images, each one with a single damage, correctly predicted by the CNN. Original images on the top line, and fusion with their respective activation maps for the predicted class on the bottom line.

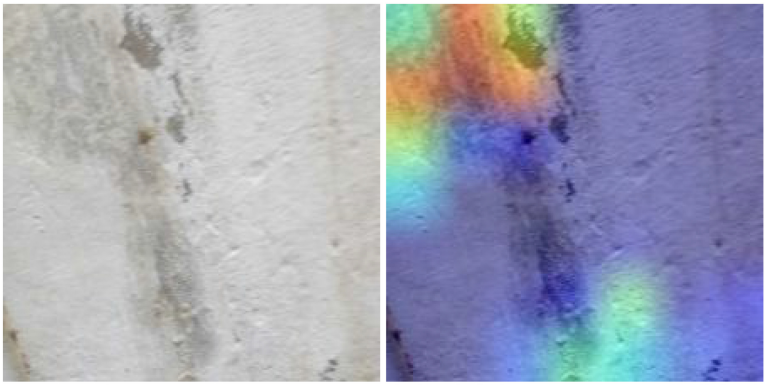


(a) Efflorescence and Crack image

(b) Grad-Cam for Efflorescence

(c) Grad-Cam for Crack

Fig. 8. An example of an image with two types of damages, correctly predicted by the CNN as containing Efflorescence along with a Crack, respectively, with the probability of 93% and 100%. (8(a)) illustrates the original picture, (8(b)) illustrates Grad-Cam for the label Efflorescence, and (8(c)) for the label Crack.



(a) Efflorescence image (b) Grad-cam for Efflorescence

Fig. 9. An illustration of a misclassified image with Efflorescence damage. Predicted probability for Efflorescence is 42% and less than 10% for the other types of damage.



(a) [9%,1%,25%,0%] (b) [24%,8%,7%,4%] (c) [5%,12%,13%,1%]

Fig. 10. An example of three images correctly predicted as damage-free and their respective probabilities following this order [Efflorescence, Spall, Crack, Mold].

in Figure 11(a), and the GRAD-CAM for this type of damage in Figure 11(b). In the latter example, GRAD-CAM shows that the CNN focused on a region without any damage and illustrates one of its limits. However, it can be noticed that the probability of prediction of the damage is only of 57%, which is not too high.

*5.1.6 Performance Evaluation Varying Wall Distance and Viewing Angle.* Test2 has been collected with variations in wall distance and viewing angle. Results are illustrated in Table 5. Compared to the AUC results on Test1, it can be noticed a slightly lower performance especially for the Efflorescence damage and Spall. Getting too far from the wall makes the CNN predict lower probabilities of damages occupying small parts of the frame. However, these results are still good, especially when taking into consideration that the objective of the CNN is to help users to take better images. An example of an Efflorescence damage with three variations in distance from the wall and two variations in the angles of view is illustrated in Figure 12. It can be noticed that the prediction probability is higher in the closest picture, as the damage is occupying a big part of the image (Figure 12(a)). As we are moving distant from the wall, the damage is occupying a smaller part of the image, and the probability is lower (Figure 12(b) and 12(c)). Being too far away from the wall can lead to indiscernible damage, even for an expert. The CNN should encourage users to get as close as possible to the damage by displaying higher probabilities.



(a) [57%,1%,3%,2%] (b) GRAD-CAM

Fig. 11. An example of an image with no damage that has been wrongly predicted as containing Efflorescence with 57% probability and its GRAD-CAM for the same label.



(a) Near distance [100%,0%,1%,0%] (b) Medium distance [94%,0%,0%,0%] (c) Far distance [60%,1%,1%,0%] (d) Right angular view [96%,0%,0%,0%] (e) Left angular view [85%,0%,6%,1%]

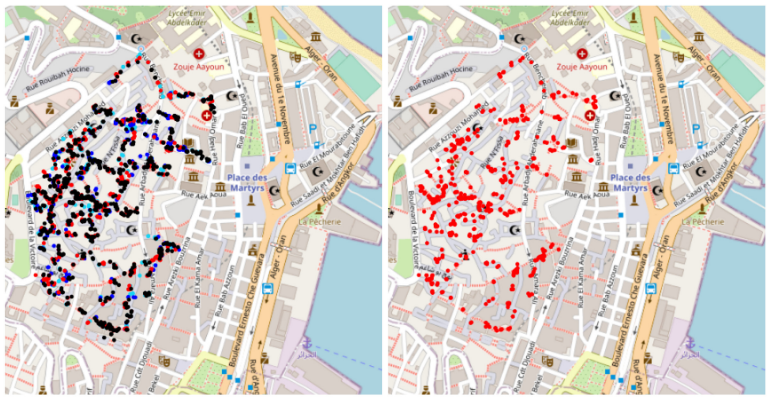
Fig. 12. An example from Test2 of an Efflorescence damage photographed from near 12(a), medium 12(b), and a far distance 12(c), and also with two angular variations (12(d), and 12(e)). Original pictures are illustrated in the top line and their respective GRAD-CAM for Efflorescence damage on the bottom. Respective predicted probabilities by the CNN are represented under each picture following this order [Efflorescence, Spall, Crack, Mold].

Based on these evaluation results, it can be noticed that the CNN is able to perform well, despite several constraints, such as the relatively small size of both the CNN and the dataset, the low resolution of the images, the variation of either the distance or the angle of view, and especially the overlapping damages.

## 5.2 Geographic Mapping

Finally, after evaluating the CNN, a set of 3,208 images containing location data has been used for a geographic mapping of surface damages.

Geographic mapping of these surface damages has been done by extracting the GPS information of each image from its EXIF metadata. Date and time of each picture, type of surface damage, latitude, and longitude characteristics have been considered. Plotly library [34] has been used to plot these characteristics on a map, giving a global idea of the distribution of each surface damage. Information of each damage has been also available when passing the mouse over it. An illustration of this geographic mapping is given in Figure 13.


Zouje Ayoun
Place des Martyrs
Rue Rouiguh Hocine
Avenue du 1e Novembre
Rue Ali Khodja
Rue Ahmed Bouzrina
Rue Ben Azzouz
Rue Ben Mahmoud
Rue Ali Boumendjel
Rue Ben Ali Chérif
Rue Ben Mhidi Larbi
Rue Ait Ameur Moh
Rue Ait Ahmed Slimane
Rue Ait Ali Ahmed
Rue Ait Aissa Amar
Rue Ait Aissa Mimoun
Rue Ait Ouakli Mohand
Rue Ait Ouali Mohand
Rue Ait Ouali Said
Rue Ait Ouali Slimane
Rue Ait Ouali Youcef
Rue Ait Ouali Youcef
Rue Ait Ouali Youcef

In particular, we will focus on incentives related to entertainment and to the exchange of services (i.e., based on mutual benefit).

## 7 ONLINE RESOURCES

Our CNN and the test1 dataset are available following the URL: https://data.mendeley.com/datasets/zwshr65gnc. They can be freely used to assess and compare our solution.

## REFERENCES


[1] Ikhlas Abdel-Qader, Osama Abudayyeh, and Michael E. Kelly. 2003. Analysis of edge-detection techniques for crack identification in bridges. *J. Comput. Civil Eng.* 17, 4 (2003), 255–263. DOI: https://doi.org/10.1061/(ASCE)0887-3801(2003)17:4(255)

[2] Mohamed E. H. Abdi, Qais Naili, Mahdi Habbache, Bilhadj Said, Abdelhak Boumenir, Taouba Douibi, Dahlia Djermane, and Sid-Ahmed Berrani. 2022. Effectively detecting left bundle branch block false defects in myocardial perfusion imaging (MPI) with a convolutional neural network (CNN). *Stud. Health Technol. Inform.* 289 (2022), 216–219.

[3] Firoj Alam, Muhammad Imran, and Ferda Ofli. 2017. Image4Act: Online social media image processing for disaster response. In *IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*. 601–604.

[4] Khaled Bayoudh, Fayçal Hamdaoui, and Abdellatif Mtibaa. 2021. Transfer learning based hybrid 2D-3D CNN for traffic sign recognition and semantic road detection applied in advanced driver assistance systems. *Appl. Intell.* 51, 1 (2021), 124–142.

[5] Simone Bianco, Remi Cadene, Luigi Celona, and Paolo Napoletano. 2018. Benchmark analysis of representative deep neural network architectures. *IEEE Access* 6 (2018), 64270–64277. DOI: https://doi.org/10.1109/access.2018.2877890

[6] Gavin C. Cawley and Nicola L. C. Talbot. 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. *J. Mach. Learn. Res.* 11, 70 (2010), 2079–2107. Retrieved from http://jmlr.org/papers/v11/cawley10a.html.

[7] Young-Jin Cha, Wooram Choi, and Oral Büyüköztürk. 2017. Deep learning-based crack damage detection using convolutional neural networks. *Comput.-aid. Civil Infrast. Eng.* 32, 5 (2017), 361–378. DOI: https://doi.org/10.1111/mice.12263

[8] Young-Jin Cha, Wooram Choi, Gahyun Suh, Sadegh Mahmoudkhani, and Oral Büyüköztürk. 2018. Autonomous structural visual inspection using region-based deep learning for detecting multiple damage types. *Comput.-aid. Civil Infrast. Eng.* 33, 9 (2018), 731–747. DOI: https://doi.org/10.1111/mice.12334

[9] François Chollet. 2017. *Transfer Learning and Fine-tuning*. Tensorflow. Retrieved from https://www.tensorflow.org/tutorials/images/transfer_learning.

[10] Yumin Du, Wenwu Chen, Kai Cui, Jingke Zhang, Zhuo Chen, and Qiyong Zhang. 2020. Damage assessment of earthen sites of the Ming Great Wall in Qinghai Province: A comparison between support vector machine (SVM) and BP neural network. *J. Comput. Cult. Herit.* 13, 2 (May 2020). DOI: https://doi.org/10.1145/3376120

[11] Cao Vu Dung and Le Duc Anh. 2019. Autonomous concrete crack detection using deep fully convolutional neural network. *Autom. Construct.* 99 (2019), 52–58. DOI: https://doi.org/10.1016/j.autcon.2018.11.028

[12] Tom Fawcett. 2006. An introduction to ROC analysis. *Patt. Recog. Lett.* 27, 8 (2006), 861–874. DOI: https://doi.org/10.1016/j.patrec.2005.10.010

[13] Marco Fiorucci, Marina Khoroshiltseva, Massimiliano Pontil, Arianna Traviglia, Alessio Del Bue, and Stuart James. 2020. Machine learning for cultural heritage: A survey. *Patt. Recog. Lett.* 133 (2020), 102–108. DOI: https://doi.org/10.1016/j.patrec.2020.02.017

[14] Tadayoshi Fushiki. 2011. Estimation of prediction error by using K-fold cross-validation. *Statist. Comput.* 21, 2 (1 Apr. 2011), 137–146. DOI: https://doi.org/10.1007/s11222-009-9153-8

[15] Spencer Gibb, Hung Manh La, and Sushil Louis. 2018. A genetic algorithm for convolutional network structure optimization for concrete crack detection. In *IEEE Congress on Evolutionary Computation (CEC'18)*. 1–8. DOI: https://doi.org/10.1109/CEC.2018.8477790

[16] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2015. Deep residual learning for image recognition. *CoRR* abs/1512.03385 (2015).

[17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR'16)*. 770–778. DOI: https://doi.org/10.1109/CVPR.2016.90

[18] Mojtaba Heidari, Mehdi Torabi-Kaveh, Carlos Chastre, Marco Ludovico-Marques, Hassan Mohseni, and Hossein Akefi. 2017. Determination of weathering degree of the Persepolis stone under laboratory and natural conditions using fuzzy inference system. *Construct. Build. Mater.* 145 (2017), 28–41. DOI: https://doi.org/10.1016/j.conbuildmat.2017.03.230

[19] Centre UNESCO World Heritage. 2021. *UNESCO World Heritage Centre - State of Conservation (SOC 2021) Kasbah of Algiers (Algeria)*. UNESCO. Retrieved from https://whc.unesco.org/en/soc/4230/.

[20] Andrew G. Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. 2017. MobileNets: Efficient convolutional neural networks for mobile vision applications. *CoRR* abs/1704.04861 (2017).

[21] Gao Huang, Zhuang Liu, and Kilian Q. Weinberger. 2016. Densely connected convolutional networks. *CoRR* abs/1608.06993 (2016).

[22] jdariasl. 2021. Global Multi-label Confusion Matrix. Retrieved from https://github.com/jdariasl/global_multiLabel_confusion_matrix.

[23] Damian Kiliszek and Krzysztof Kroszczyński. 2020. Performance of the precise point positioning method along with the development of GPS, GLONASS and Galileo systems. *Measurement* 164 (2020), 108009. DOI: https://doi.org/10.1016/j.measurement.2020.108009

[24] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E. Hinton. 2012. ImageNet classification with deep convolutional neural networks. In *Advances in Neural Information Processing Systems*, Vol. 25. Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf.

[25] Damir Krstinić, Maja Braović, Ljiljana Šerić, and Dunja Božić-Štulić. 2020. Multi-label classifier performance evaluation with confusion matrix. *Comput. Sci. Inf. Technol.* 10, 8 (2020).

[26] Pakhee Kumar, Ferda Ofli, Muhammad Imran, and Carlos Castillo. 2020. Detection of disaster-affected cultural heritage sites from social media images using deep learning techniques. *J. Comput. Cult. Herit.* 13, 3 (Aug. 2020). DOI: https://doi.org/10.1145/3383314

[27] S. Lawrence, C. L. Giles, Ah Chung Tsoi, and A. D. Back. 1997. Face recognition: A convolutional neural-network approach. *IEEE Trans. Neural Netw.* 8, 1 (1997), 98–113. DOI: https://doi.org/10.1109/72.554195

[28] Y. LeCun, B. Boser, J. S. Denker, D. Henderson, R. E. Howard, W. Hubbard, and L. D. Jackel. 1989. Backpropagation applied to handwritten zip code recognition. *Neural Computat.* 1, 4 (1989), 541–551. DOI: https://doi.org/10.1162/neco.1989.1.4.541

[29] Shengyuan Li, Xuefeng Zhao, and Guangyi Zhou. 2019. Automatic pixel-level multiple damage detection of concrete structure using fully convolutional network. *Comput.-aid. Civil Infrast. Eng.* 34, 7 (2019), 616–634.

[30] Mayank Mishra. 2021. Machine learning techniques for structural health monitoring of heritage buildings: A state-of-the-art review and case studies. *J. Cultur. Herit.* 47 (2021), 227–245. DOI: https://doi.org/10.1016/j.culher.2020.09.005

[31] Marie-Francine Moens, Gareth J. F. Jones, Saptarshi Ghosh, Debasis Ganguly, Tanmoy Chakraborty, and Kripabandhu Ghosh. 2018. WWW'18 workshop on exploitation of social media for emergency relief and preparedness: Chairs' welcome & organization. In *the Web Conference (WWW'18)*. International World Wide Web Conferences Steering Committee, 1609–1611. DOI: https://doi.org/10.1145/3184558.3192321

[32] Rupa Patel and Anita Chaware. 2020. Transfer learning with fine-tuned MobileNetV2 for diabetic retinopathy. In *International Conference for Emerging Technology (INCET'20)*. 1–4. DOI: https://doi.org/10.1109/INCET49848.2020.9154014

[33] Ramani Perumal and Subbiah Bharathi Venkatachalam. 2021. Non invasive detection of moss and crack in monuments using image processing techniques. *J. Amb. Intell. Human. Comput.* 12, 5 (1 May 2021), 5277–5285. DOI: https://doi.org/10.1007/s12652-020-02006-x

[34] Plotly. 2022. *Mapbox Layers*. Plotly. Retrieved from https://plotly.com/python/mapbox-layers/.

[35] Haran Pragalath, Sankarasrinivasan Seshathiri, Harsh Rathod, Balasubramanian Esakki, and Rishi Gupta. 2018. Deterioration assessment of infrastructure using fuzzy logic and image processing algorithm. *J. Perform. Construct. Facil.* 32, 2 (2018), 04018009. DOI: https://doi.org/10.1061/(ASCE)CF.1943-5509.0001151

[36] Payam Refaeilzadeh, Lei Tang, and Huan Liu. 2009. *Cross-validation*. Springer US, Boston, MA, 532–538. DOI: https://doi.org/10.1007/978-0-387-39940-9_565

[37] Shaoqing Ren, Kaiming He, Ross B. Girshick, and Jian Sun. 2015. Faster R-CNN: Towards real-time object detection with region proposal networks. *CoRR* abs/1506.01497 (2015).

[38] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael S. Bernstein, Alexander C. Berg, and Li Fei-Fei. 2014. ImageNet large scale visual recognition challenge. *CoRR* abs/1409.0575 (2014).

[39] Mark Sandler, Andrew G. Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. 2018. Inverted residuals and linear bottlenecks: Mobile networks for classification, detection and segmentation. *CoRR* abs/1801.04381 (2018).

[40] Ramprasaath R. Selvaraju, Abhishek Das, Ramakrishna Vedantam, Michael Cogswell, Devi Parikh, and Dhruv Batra. 2016. Grad-CAM: Why did you say that? Visual explanations from deep networks via gradient-based localization. *CoRR* abs/1610.02391 (2016).

[41] Connor Shorten and Taghi M. Khoshgoftaar. 2019. A survey on image data augmentation for deep learning. *J. Big Data* 6, 1 (6 July 2019), 60. DOI: https://doi.org/10.1186/s40537-019-0197-0

[42] Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556* (2014).

[43] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott E. Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. 2014. Going deeper with convolutions. *CoRR* abs/1409.4842 (2014).

[44] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jonathon Shlens, and Zbigniew Wojna. 2015. Rethinking the inception architecture for computer vision. *CoRR* abs/1512.00567 (2015).

[45] Mingxing Tan and Quoc V. Le. 2021. EfficientNetV2: Smaller models and faster training. *CoRR* abs/2104.00298 (2021).

[46] Keras Team. 2021. *Keras Documentation: Keras Applications*. Keras. Retrieved from https://keras.io/api/applications/.

[47] Niannian Wang, Qingan Zhao, Shengyuan Li, Xuefeng Zhao, and Peng Zhao. 2018. Damage classification for masonry historic structures using convolutional neural networks based on still images. *Comput.-aid. Civil Infrast. Eng.* 33, 12 (2018), 1073–1089. DOI: https://doi.org/10.1111/mice.12411

[48] Niannian Wang, Xuefeng Zhao, Linan Wang, and Zheng Zou. 2019. Novel system for rapid investigation and damage detection in cultural heritage conservation based on deep learning. *J. Infrast. Syst.* 25, 3 (2019), 04019020. DOI: https://doi.org/10.1061/(ASCE)IS.1943-555X.0000499

[49] Niannian Wang, Xuefeng Zhao, Peng Zhao, Yang Zhang, Zheng Zou, and Jinping Ou. 2019. Automatic damage detection of historic masonry buildings based on mobile deep learning. *Autom. Construct.* 103 (2019), 53–66. DOI: https://doi.org/10.1016/j.autcon.2019.03.003

[50] Niannian Wang, Xuefeng Zhao, Zheng Zou, Peng Zhao, and Fei Qi. 2020. Autonomous damage segmentation and measurement of glazed tiles in historic buildings via deep learning. *Comput.-aid. Civil Infrast. Eng.* 35, 3 (2020), 277–291. DOI: https://doi.org/10.1111/mice.12488

[51] Qian Xiang, Xiaodan Wang, Rui Li, Guoling Zhang, Jie Lai, and Qingshuang Hu. 2019. Fruit image classification based on MobileNetV2 with transfer learning technique. In *3rd International Conference on Computer Science and Application Engineering (CSAE'19)*. Association for Computing Machinery, New York, NY. DOI: https://doi.org/10.1145/3331453.3361658

[52] Matthew D. Zeiler and Rob Fergus. 2014. Visualizing and understanding convolutional networks. In *Computer Vision – ECCV 2014*, David Fleet, Tomas Pajdla, Bernt Schiele, and Tinne Tuytelaars (Eds.). Springer International Publishing, Cham, 818–833.

[53] Xinglin Zhang, Zheng Yang, Wei Sun, Yunhao Liu, Shaohua Tang, Kai Xing, and Xufei Mao. 2016. Incentives for mobile crowd sensing: A survey. *IEEE Commun. Surv. Tutor.* 18, 1 (2016), 54–67. DOI: https://doi.org/10.1109/COMST.2015.2415528

[54] Barret Zoph, Vijay Vasudevan, Jonathon Shlens, and Quoc V. Le. 2017. Learning transferable architectures for scalable image recognition. *CoRR* abs/1707.07012 (2017).

[55] Kivanc Zorlu. 2008. Description of the weathering states of building stones by fractal geometry and fuzzy inference system in the Olba ancient city (Southern Turkey). *Eng. Geol.* 101, 3 (2008), 124–133. DOI: https://doi.org/10.1016/j.enggeo.2008.04.005

[56] Zheng Zou, Xuefeng Zhao, Peng Zhao, Fei Qi, and Niannian Wang. 2019. CNN-based statistics and location estimation of missing components in routine inspection of historic buildings. *J. Cultur. Herit.* 38 (2019), 221–230. DOI: https://doi.org/10.1016/j.culher.2019.02.002

Received 6 April 2022; revised 2 July 2022; accepted 9 September 2022